JOS THOMAS

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Scientific Computing • Original Algorithms • C++/MPI • GPU/HPC

EDUCATION

DePaul University | Chicago, IL Expected June 2027

B.S. in Computer Science (Software Development concentration) | GPA: **4.00/4.00**

Oakton College | Des Plaines, IL 2025

Associate in Arts, Graduated with Honors

RESEARCH EXPERIENCE

Los Alamos National Laboratory | Theoretical Division, T-1 2025–Present

Invented ALAMOS and ALAMOS-0 | First-author LANL–Sandia paper submitted to JCP

Research project support: U.S. Department of Energy, Office of Basic Energy Sciences; Los Alamos National Laboratory's Laboratory Directed Research and Development (LDRD) program.

Scientific Computing Research Intern Jun 2026–Present

On-site Jun–Sep 2026; part-time remote since Oct 2026

Research mentors: Anders M. N. Niklasson and Anthony P. Baldo

- **ALAMOS-0:** Developed a preconditioning approximation that **eliminates all corrective Krylov iterations**; demonstrated **stable 104,640-atom, 100-ps NVE water dynamics (200,000 steps)** in SEDACS on one NVIDIA A100, with charge-dependent calculations accounting for only **~8.4% of runtime.**
- **LAMMPS:** Implemented Shadow XL-BOMD (Levels 0/1) with ALAMOS-0 in C++/MPI; passed multi-rank, restart, and charge-accuracy tests; now validating stability and scalability for integration into open-source LAMMPS.
- **PPCG**: Developed a GPU solver with a rank-one Schur-complement charge-conservation correction; reduced charge-dependent runtime by 39% versus the original diagonally preconditioned GMRES solver (52,320 atoms).
- **ALAMOS**: Invented a multiscale preconditioner; with PPCG, reduced charge-dependent runtime by 61% versus GMRES (52,320 atoms) and reduced solver iterations nearly 13-fold in a separate benchmark.
- **AMOS:** Developed a novel method to accelerate global charge-response calculations; now evaluating numerical accuracy, long-term stability, and scalability.
- **VALLES:** Created an FFT-free long-range electrostatics method and derived energy-consistent forces; validated against Ewald and demonstrated stable 104,640-atom, 100-ps dynamics on one A100 GPU.
- **Physical validation:** Implemented GJ-I Langevin dynamics; 1.1-ns simulations at 100 K and 300 K reproduced theoretical temperature distributions. IR spectra matched converged-charge references; a 300 K reference-charge spectrum from an ALAMOS-0 trajectory reproduced characteristic experimental water bands.

Undergraduate Researcher | Remote Oct 2025–Jun 2026

- Developed constrained charge-equilibration models for shadow MD; analyzed charge-response Jacobians, residuals, energy conservation, and stability, establishing the foundation for later algorithm development.

Machine Learning & Scientific Computing Research Intern | Remote May–Aug 2025

- Selected as 1 of 11 students nationwide for a Texas A&M–LANL research program; developed a fourth-generation machine-learned interatomic potential for 2D perovskites with flexible charges.

PREPRINTS & MANUSCRIPTS

Thomas, J., Baldo, A. P., Corrigan Grove, R. A., Kaymak, M. C., Finkelstein, J., Goff, J., Kulichenko, M., Tretiak, S., Lubbers, N., Athavale, V., Negre, C. F. A., Wall, M., and Niklasson, A. M. N. *Shadow Molecular Dynamics for General Nonlinear Flexible-Charge Models.* Los Alamos National Laboratory, LA-UR-26-29564 (2026). **arXiv:2611.03499**↗. Submitted to *The Journal of Chemical Physics* (November 2026).

Thomas, J., Baldo, A. P., and Niklasson, A. M. N. First-author ALAMOS methods manuscript (in preparation, 2026).

Thomas, J., et al. Manuscript on induced dipoles and D4 dispersion (in preparation, 2026).

HONORS & AWARDS

AMD Foundation | Awarded funding for SC26, the international HPC conference 2026
Los Alamos National Laboratory Intern Award | Recognition for scientific research contributions 2026
Lam Research | Undergraduate tuition award 2026
Cuneo Foundation Scholarship | $10,000; 1 of 4 DePaul recipients for academics, service, and potential 2026
Upsilon Pi Epsilon | Invited to ACM/IEEE-CS-endorsed international computing honor society 2026
DePaul Transfer Scholarship | $20,000/year academic merit scholarship 2025–27
DePaul Dean's List | Recognized for academic excellence in every eligible term 2025–26
Oakton College Educational Foundation Scholarship | Undergraduate scholarship award 2024–25
Oakton President's Scholar | Perfect 4.00 term GPA across 5 terms 2020–25

RESEARCH PRESENTATIONS

"GPU-Accelerated Shadow Molecular Dynamics Using Machine-Learned Interatomic Potentials with Long-Range Charge Equilibration"
LANL Student Symposium | Poster presentation Aug 4, 2026
LANL Theoretical Division (T-1) | Lightning talk Jul 30, 2026

"Development of Fourth-Generation Machine-Learned Potential with Long-Range Flexible Charges for Two-Dimensional CsPbI Perovskites"
Texas A&M University | Research presentation ↗ Jul 23, 2025
Texas A&M University | Research progress presentation Jun 25, 2025

TECHNICAL SKILLS

Programming & HPC: C++, Python, Java, MATLAB, SQL; MPI, GPU computing, parallel programming, Linux, Git
Numerical & Simulation Methods: Numerical linear algebra, Krylov solvers (CG, GMRES), preconditioning, charge equilibration, molecular dynamics, numerical electrostatics
Scientific Computing: LAMMPS, PyTorch, NumPy, SciPy, ASE, pymatgen, scikit-learn, pandas, Matplotlib

ADDITIONAL PROFESSIONAL EXPERIENCE

How-To Geek | Linux Specialist Dec 2024–May 2025
- Tested and troubleshot Linux software; published articles reaching 120,000+ readers and earned Top Contributor recognition.

Keats Manufacturing Co. | Software Engineering Intern Sep–Dec 2024
- Developed Python software to automate manufacturing measurement validation and out-of-tolerance detection, reducing manual review and accelerating identification of production issues.

Keats Manufacturing Co. | Quality Control Technician Jun 2016–Sep 2024
- Analyzed manufacturing data and prepared quality reports while working night shifts through college.

SELECTED SOFTWARE PROJECT

Multi-Agent Mobile QA System | Python, Google ADK, Gemini, ADB Code ↗
- Architected a Supervisor–Planner–Executor multi-agent Android QA system that executes natural-language tests via ADB, verifies UI state, and generates auditable screenshot and JSONL traces.

MENTORING, OUTREACH & SERVICE

Independent Programming Tutor 2025–Present
- Mentored 14 high-school students in programming and debugging through Scratch game development.

Citadel at Saint Benedict | Volunteer Jun 2015–Present
- Assist residents with meals and mobility, support staff, and train new volunteers (11+ years; ~8 hours/week).

Summer Camp Counselor / Head Counselor 2019–2023
- Mentored campers for 5 summers; supervised counselors and coordinated activities as head counselor in 2023.